

- ① A network shows an open-circuit voltage of 9V and a short-circuit current of 1.5A at its terminals. Find R_L in Ω if the terminal voltage for a 6Ω load.

⇒ Open Circuit Voltage, $V_{OC} = 9\text{V}$

Short Circuit Current, $I_N = I_{SC} = 1.5\text{A}$

Load Resistance, $R_L = 6\Omega$

To find R_L ,

From Thevenin-Norton Relationship, $R_{th} = \frac{V_{OC}}{I_N} = \frac{9}{1.5} = 6\Omega$.

$$\therefore R_N = R_{th} = 6\Omega$$

Terminal voltage for 6Ω load,
Using Thevenin equivalent circuit,

$$V_L = \frac{V_{OC} R_L}{R_{th} + R_L} = 9 \left(\frac{6}{6+6} \right) = 3.6\text{V}$$

2. A 100mH inductor & a 47nF capacitor are placed in series, find the frequency at which their reactances are equal.

⇒ Given,

$$L = 100\text{mH} = 0.1\text{H}$$

$$C = 47\text{nF} = 47 \times 10^{-9}\text{F}$$

At equal reactance, $X_L = X_C \Rightarrow \omega L = \frac{1}{\omega C}$

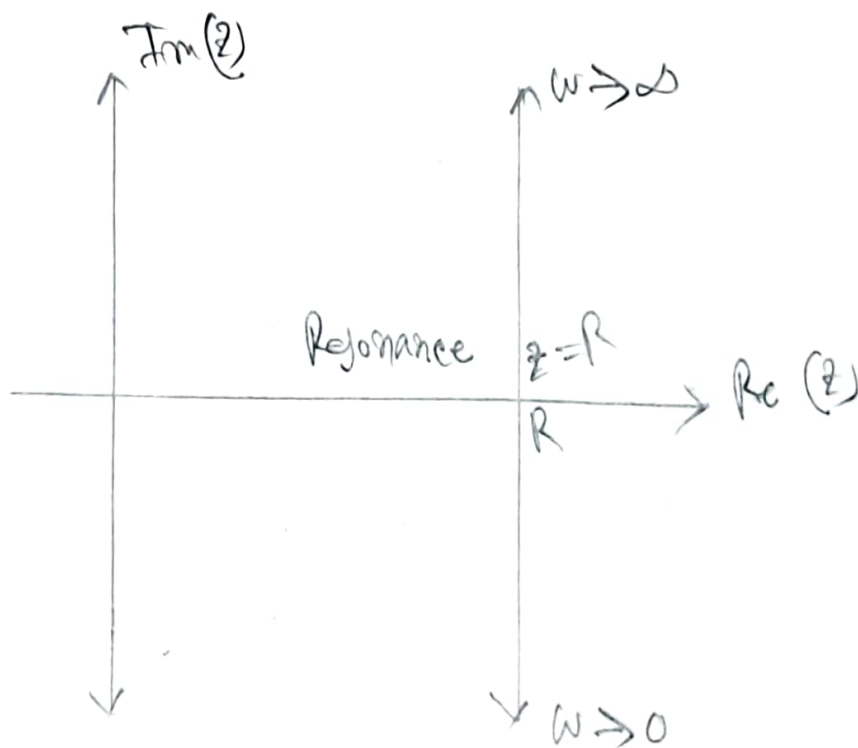
$$\therefore \omega_0 = \frac{1}{\sqrt{LC}}$$

$$\Rightarrow \omega_0 = \frac{1}{\sqrt{(0.1)(47 \times 10^{-9})}} = 461.9\text{ rad/s.}$$

$$\text{Now, } f_0 = \frac{\omega_0}{2\pi} = \frac{461.9}{2\pi} = 73.5\text{ Hz.}$$

Impedance locus,

The locus is a vertical straight line at $\text{Re}(z) = R$.



③ An inductor voltage waveform is +20V for 3ms & - U_2 for the remaining 2ms of a 5ms period. Find U_2

⇒ Given,

Total period = 5ms,

At Steady State operation, $\int_0^T U_L(t) dt = 0$

∴ Positive & negative volt-seconds must be equal.

Apply Volt-Second balance,

$$(20)(3 \times 10^{-3}) + (-U_2)(2 \times 10^{-3}) = 0$$

$$\Rightarrow 60 \times 10^{-3} - 2U_2 \times 10^{-3} = 0$$

$$\Rightarrow 60 = 2U_2 \Rightarrow \boxed{U_2 = 30V}$$

∴ wave form of Inductor +20V for 3ms & -30V for 2ms.

④ Sketch the impedance locus of a series RLC circuit as ω is swept from 0 to ∞ , & indicating the sketch at which the locus crosses the real axis.

⇒ For a series RLC circuit,

$$Z = R + j\omega L - j/\omega C$$

$$\therefore X_L = \omega L \text{ \& } X_C = 1/\omega C$$

$$\therefore Z = R + j(\omega L - 1/\omega C)$$

$$\therefore \text{Re}(Z) = R \text{ \& } \text{Im}(Z) = \omega L - 1/\omega C$$

At $\omega = 0$, $X_L = 0$ but $X_C \rightarrow \infty \therefore \text{Im}(Z) \rightarrow -\infty$

So the locus starts at $R - j\infty$

At resonance,

The locus crosses the real axis when $x_L = x_C$

$$\therefore \omega_{0L} = \omega_{0C} \Rightarrow \omega_0 = \frac{1}{\sqrt{LC}}$$

At this point, $x_L = x_C = 0$

$$\Rightarrow f_0, Z = R,$$

At $\omega \rightarrow \infty$, $x_L \rightarrow \infty$ & $x_C \rightarrow 0 \therefore \text{Im}(Z) \rightarrow +\infty$

So the locus goes toward: $R + j\infty$

⑤ Using Euler's Formula, express $u(t) = 20 \cos(314t + 30^\circ)$ as a phasor & state the frequency in Hz.

Given,

$$u(t) = 20 \cos(314t + 30^\circ)$$

Standard sinusoidal form, $u(t) = U_m \cos(\omega t + \phi)$

Here, $U_m = 20$, $\omega = 2\pi f$ rad/s, $\phi = 30^\circ$

$\therefore$ phasor is $20 \angle 30^\circ$ $U = U'$

$$U_{rms} = \frac{20}{\sqrt{2}} = 14.14 \text{ V}, \text{ then, } U' = 14.14 \angle 30^\circ \text{ V}$$

we know,

$$\omega = 2\pi f \Rightarrow 314 = 2\pi f \Rightarrow f = 50 \text{ Hz}$$

$$\therefore f \approx 50 \text{ Hz.}$$

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